

ADDENDUM 2 to KIMI_LPW_REVIEW_VERDICT.md — custody and diagnostic corrections (2026-09-13)

The sealed verdict (body d399e28378d177f88d08032c81f1573f35c9af2b600b9b775ebfe40f555cbbad) and the sealed Addendum 1 (body ca58855f8344c1c1...) remain unchanged in the record. This second addendum repairs two defects identified by the author-side's Return-03 package (zip sha256 fb81f8c1774d851775a025ac08750fa9e98e7b04d0e6844e7aebfd14456de2c7, 306,334 B). The LPW endorsement itself (all six interfaces PASS WITHIN SCOPE at cf58f72e...e1a5) is untouched.

1. Malformed whole-file digest (custody defect, owned)

Addendum 1 §“Sealed” and the README printed the verdict's whole-file hash as 04bcbdf2013d83b113c18518bd101bb8c1a96 — 66 hexadecimal characters; a SHA-256 digest has 64. This was a transcription slip in the addendum drafting, not a hidden-file issue. Recomputed from the actual bytes of REVIEW_LPW/KIMI_LPW_REVIEW_VERDICT.md (9,219 B):

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whole-file SHA-256 = 04bcbdf2013d83b149a8933a332a078c303af347335ee40f154f37ba195ad448
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(the printed prefix 04bcbdf2013d83b1 is correct). The README_BUNDLE.md hash line has been corrected in place (it is a packaging file, not a sealed transcript) and the bundle MANIFEST.sha256 rebuilt; the declared body hash d399e28378d177f88d08032c81f1573f35c9af2b600b9b775ebfe40f555cbbad is unaffected and remains governed by the stated extraction rule (hash the exact byte prefix before the unique marker line, preserving all preceding bytes including blank lines — never a trim-and-append operation).

2. The $a - 3$ sign (diagnostic defect, owned)

My earlier quick computation printed $a_4 - 3 \approx -4.0219e-124$. That is wrong in sign and magnitude. Cause identified (mine): a wrong fourth-derivative formula in the quick check ($a - 24 \cdot \Sigma n^4 \cdot e^{\{-288n^2\}}$ form instead of the correct fourth Hermite form $(24n) - 6(24n)^2 + 3$). The author-side's exact identity is correct:

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a_4 - 3 = 2 \cdot \Sigma_{n \geq 1} (331776n^4 - 3456n^2) \cdot e^{\{-288n^2\}} / (1 + 2 \cdot \Sigma_{n \geq 1} e^{\{-288n^2\}}) > 0,
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with every numerator summand strictly positive $(3456n^2(96n^2 - 1) > 0$ for $n \geq 1$). Independently recomputed at 150 dps with the correct Hermite form:

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a_4 - 3 = +5.5019488254071504413555...e-120 (positive),
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agreeing with the author-side's value 5.50194882540715044135552121887e-120 to 1.8e-150. Their exponential enclosure $5.50194e-120 < a - 3 < 5.50196e-120$ is consistent. The $a - 1$ diagnostic ($-9.6525e-123$) stands as previously recorded (its sign is negative and correct). The proved endpoint floor $\lfloor \Gamma_{\{0,24\}} \rfloor > 0.124$ is unaffected (it used a much looser absolute correction bound that accommodates the corrected value).

3. Raw reviewer archive (the actual transfer)

The author-side correctly notes that PDFs describe the bundle but are not the bundle. The raw archive is now produced as a single ZIP of REVIEW_LPW (29 files + MANIFEST.sha256 + README_BUNDLE.md), with whole-file sha256 for every member and the extraction rule for the verdict's declared body hash stated in the README. It contains: the verdict and both addenda;

the lead evidence pack; the four interface reports (RA/RB/RC/RD); all interface scripts; both-mode transcripts (byte-identical for RB/RC/RD; RD's original transcripts are present as out_normal.json/out_opt.json and independently reproduction-confirmed by a lead rerun matching the reported hash prefix 45f4d471...); exit receipts; ENVIRONMENT_LOCK.json (python 3.12.12, mpmath 1.3.0, numpy 2.2.5, sympy 1.14.0, scipy 1.16.2 for W2_sanity). Packaging note (unchanged from README): RA did not persist a standalone script to the shared tree; its transcript is embedded in RA_R1_R6.md, and per the author's own instruction no manufactured "recovered script" is included — a reconstruction would carry a new identity and an accurate exposure record if ever produced.

4. The Return-03 quantitative candidate

The new explicit-constants candidate ($r_* = 10^{-28}$, $c_* = 10^{-1235}$: $1 - q(r, 6/5) \cdot 10^{-1235} \cdot r^3$ for $0 < r \leq 10^{-28}$) is under independent KIMI review at its own Q1-Q7 targets. It does not inherit the qualitative endorsement. Preliminary verification (lead, already complete): the covariance perturbation algebra $\|\Gamma_r - \Gamma_0\|_{op} \leq 2\|V_0\|\|V_r - V_0\| + \|V_r - V_0\|^2 \leq 28r + 4r^2 \leq 32r$ is correct; the endpoint floor $31/250 > 1/8$ is verified (Sylvester minors, prior addendum); the choice $R = 1/1600$ then gives $\min(\Gamma_r) > 31/250 - 32/1600 = 13/125 > 1/10$ by correct arithmetic; the density exponent $(R_J + M)^2/(2) = (12+3)^2/(2 \cdot (1/10)) = 1125$ gives a natural floor $\sim 10^{-493}$ (the candidate's $m^* = 10^{-1129}$ is a valid but deliberately much weaker lower choice, ~ 636 orders below the natural floor); $\beta = 2^{-40}$ and $260/2^{40} > 10^{-10}$, so $c = 260 \cdot m \cdot \beta / B_3 \leq 10^{-1129-10-96} = 10^{-1235}$ as printed; and $r_* = 10^{-28} < (256K)^{-1} = (512 \cdot 10^{24})^{-1} \approx 1.95e-27$, so the Taylor radius is respected. The full Q1-Q7 dispositions (the L^2 estimates $\|V_r - V_0\| \leq 2r$ and $\|V_0\| < 7$, the mean bound $\|\Gamma_r\| < 3$, the Σ_r bounds, the full-Fourier $B_3 = 10^{96}$ and $B_4 = 10^{24}$, and the $r_* = 10^{-28} < \min(R, 1, (256K)^{-1})$ radius logic) are the review's remaining items; its verdict ships separately. A defect in the quantitative extension would not touch the qualitative LPW endorsement.

SHA-256 of this addendum body (text after this line is excluded): 55511a324250f532a28bdda860ae9604e43b22d278668cd52